Custom Notifications Management

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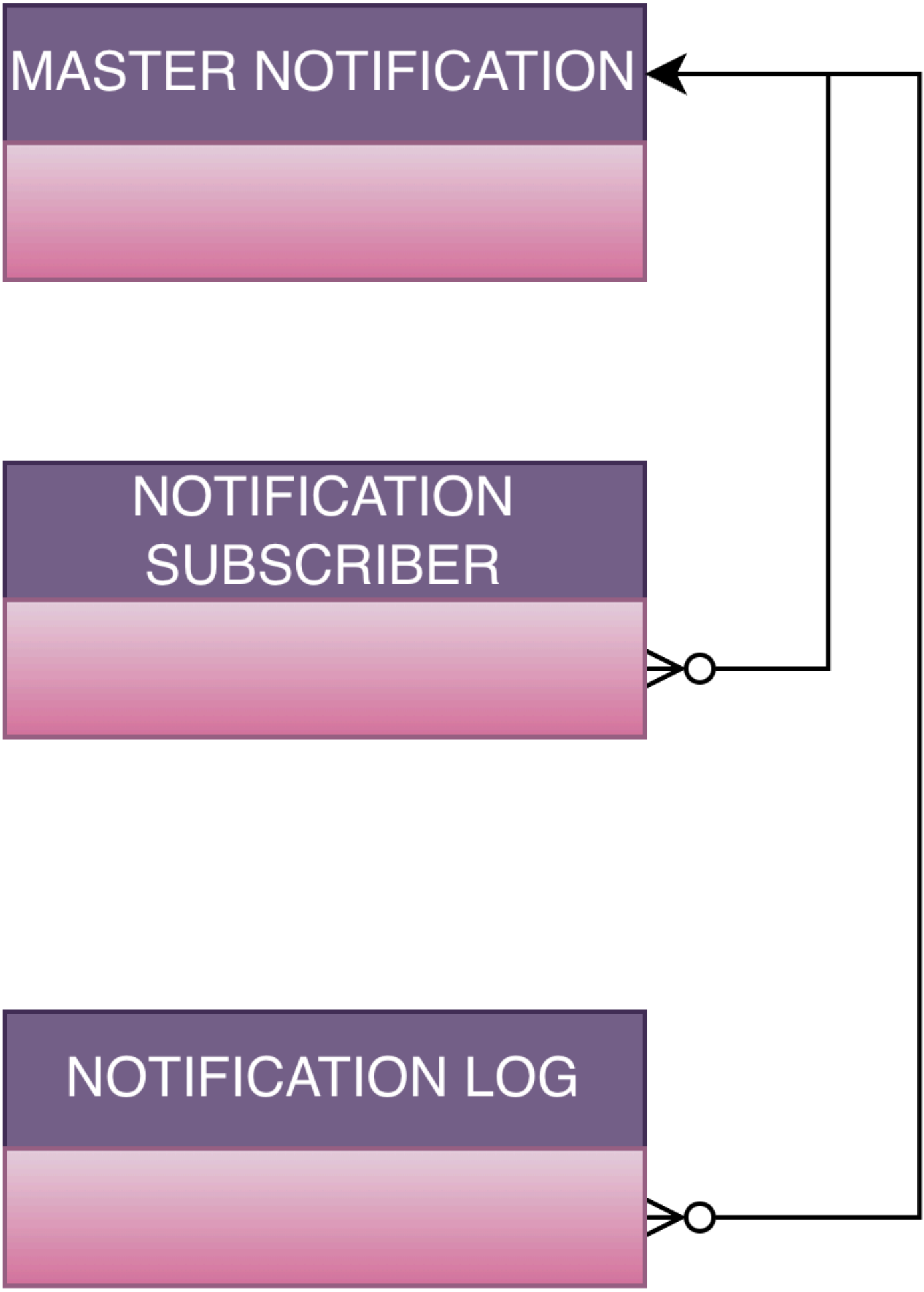

Executive Summary

The **Custom Notifications Management** solution is a metadata-driven architecture designed to handle Salesforce standard bell notifications dynamically. Instead of creating and maintaining complex, hard-coded Flows or Apex triggers for every new alert requirement, this solution provides a centralized, scalable "Rule Engine." Administrators can configure the triggering conditions, dynamic message templates, and target audiences (Users, Public Groups, or Roles) purely through data records, requiring zero new code or automations for future notifications.

Business Benefits

- **Scalability:** New notifications across any Salesforce object can be created instantly via the UI without developer intervention.
- **Maintainability:** Eliminates the proliferation of one-off Flows, significantly reducing technical debt.
- **Dynamic Content:** Supports merge fields (placeholders) in the notification subject and body to provide contextual information to the user.
- **Spam Prevention:** Includes a built-in logging system with unique identifiers to ensure duplicate notifications are not sent for the same event.

Technical Architecture & Data Model

The system relies on one standard Salesforce Setup configuration and three custom objects to separate the rule definition (the "What" and "When"), the audience definition (the "Who"), and the historical tracking.

Standard Setup Configuration

Before creating any custom objects, the standard Custom Notification Type must be enabled.

- **Component:** Custom Notification Type (Setup > Custom Notifications)
- **Notification Name:** `General Notification`
- **Notification API Name:** `WESAL_nt_generalNotification`
- **Supported Channels:** Desktop, Mobile
- **Purpose:** Generates the unique Type ID required by the Salesforce system to trigger the bell icon.

Master Notification

This object stores the business logic, triggering conditions, and the message template.

Field Label	Field Name (API)	Data Type	Description
Notification Name	`Name`	Text	Internal identifier for the rule (e.g., "Order Delivered").
Object API Name	`WESAL_fld_objectAPIName__c`	Text (100)	The API name of the triggering object (e.g., `WESAL_obj_orderRequest__c`).
Field API Name	`WESAL_fld_fieldAPIName__c`	Text (100)	The field to monitor for changes (e.g., `WESAL_fld_status__c`).
Trigger Value	`WESAL_fld_triggerValue__c`	Text (255)	The specific value that triggers the alert (e.g., `Delivered`).
Notification Title	`WESAL_fld_notificationTitle__c`	Text (255)	Title of the alert. Supports placeholders (e.g., `Order Request {Name}`).
Notification Body	`WESAL_fld_notificationBody__c`	Long Text Area	The message body. Supports placeholders (e.g., `Order {Name} is now {WESAL_fld_status__c}`).
Is Active	`WESAL_fld_active__c`	Checkbox	Toggle to enable/disable the notification rule. Default: FALSE.
Sales Org	`WESAL_fld_salesOrg__c`	Global Picklist	Filters the notification rule to apply only to a specific Sales Organization. Can be left blank if the rule applies globally across all organizations.

Notification Subscriber

This object handles the audience. It allows linking multiple recipients to a single Master Notification.

Field Label	Field Name (API)	Data Type	Description
Master Notification	`WESAL_fld_masterNotification__c`	Master-Detail	Links to the parent rule.
Recipient Type	`WESAL_fld_recipientType__c`	Picklist	Values: `User`, `Public Group`, `Role`.
Recipient ID	`WESAL_fld_recipientID__c`	Text (18)	The 18-character Salesforce ID of the User, Group, or Role
Target Record ID	`WESAL_fld_targetRecordID__c`	Text (18)	The 18-character Salesforce record ID

Notification Log

This object serves as an audit trail and an anti-spam safeguard.

Field Label	Field Name (API)	Data Type	Description
Master Notification	`WESAL_fld_masterNotification__c`	Lookup	Links to the rule that generated the log.
Target Record ID	`WESAL_fld_targetRecordId__c`	Text (18)	The ID of the record that triggered the alert (e.g., the Order Request ID).
Recipient ID	`WESAL_fld_recipientID__c`	Text (18)	The ID of the specific User who received the alert.
External ID	`WESAL_fld_externalID__c`	Text (255) (Unique, Case-Sensitive)	Concatenation to prevent duplicates.

Execution Logic (The Dispatcher)

To ensure maximum efficiency and properly handle Public Groups, Roles, and string manipulation for dynamic placeholders, the core dispatcher should be implemented using an **Apex Invocable Method** called by a simple Record-Triggered Flow.

Step-by-Step System Flow:

Event Detection (Flow): A lightweight Record-Triggered Flow on the target object (e.g., `WESAL_obj_orderRequest__c`) detects a change in a monitored field (e.g., `WESAL_fld_status__c != PriorValue`). The Flow calls the Apex Invocable Method, passing three parameters: the `RecordId`, the `Object API Name`, and the **resolved Sales Org context value** (e.g., extracting `{!$Record.WESAL_fld_account__r.WESAL_fld_salesOrganization__c}`)

2. **Rule Evaluation (Apex):** The Apex class queries active `WESAL_obj_masterNotification__c` records where:

- `WESAL_fld_objectAPIName__c` matches the passed Object name.
- The target field's new value matches the `WESAL_fld_triggerValue__c`.
- `WESAL_fld_active__c` = `TRUE`.
- `WESAL_fld_salesOrg__c` equals the Sales Org string passed by the Flow, OR `WESAL_fld_salesOrg__c` = `NULL` (for global rules).

3. **Audience Resolution (Apex):** * The system queries related `WESAL_obj_notificationSubscriber__c` records.
   - If the type is `User`, it adds the ID to the recipient list.
   - If the type is `Public Group` or `Role`, Apex executes a SOQL query on `GroupMember` or `UserRole` to extract the individual User IDs.
   - Filter By WESAL_fld_targetRecordID__c
4. **Placeholder Resolution (Apex):** The system parses the `WESAL_fld_notificationTitle__c` and `WESAL_fld_notificationBody__c`, finding tags like `{WESAL_fld_status__c}` and replacing them with the actual values from the triggering record.

5. **Deduplication Check (Apex):** For each recipient, a string is generated: `[MasterNotificationId]-[TargetRecordId]-[UserId]`. The system checks against `Notification_Log__c.External_ID__c` or handles the `DML Exception` on insert. If it already exists, the user is skipped.

6. **Dispatch (Apex):** The system instantiates the `Messaging.CustomNotification` class, setting the Type ID, Title, Body, Target Record ID, and sends it to the compiled set of User IDs.

7. **Logging (Apex):** Inserts the successful dispatches into the `WESAL_obj_notificationLog__c` object.

Deployment & Implementation Steps for the Dev Team

1. **Setup Creation:** Create the Custom Notification Type.

2. **Schema Deployment:** Create the 3 Custom Objects, fields, and relationships.
3. **Apex Development:** Write the `WESAL_cls_notificationDispatcher` class (Invocable Method) to handle audience expansion, string replacement, deduplication, and sending via `Messaging.CustomNotification`.
4. **Trigger Flow Creation:** Build the generic Record-Triggered Flow on `WESAL_obj_orderRequest__c` (and future objects) to pass the initial variables to the Apex class.
5. **Testing & QA:** Validate against single users, groups, roles, and verify that duplicate trigger events do not create multiple logs/alerts.

Record-Level Subscription UI (LWC)

To provide an intuitive user experience, a Lightning Web Component named `wesal_lwc_notificationManager` will be deployed on the Record Page of supported objects (e.g., `WESAL_obj_orderRequest__c`). This component allows users to manually subscribe themselves or others to specific notification rules for the current record.

Functional Requirements

- **Rule Selection:** A dropdown filtered by the current Object API Name to select active `WESAL_obj_masterNotification__c` records.
- **Recipient Targeting:** A dynamic search interface (Lookup) to select:
  - **Users:** Queries the `User` object.
  - **Public Groups:** Queries the `Group` object (Type = 'Regular').
  - **Roles:** Queries the `UserRole` object.
- **Context Awareness:** The component automatically captures the `recordId` from the page and populates the `WESAL_fld_targetRecordId__c` field.

Technical Specifications (Apex Controller)

The LWC will be supported by an Apex Controller (`WESAL_cls_notificationUIController`) with the following methods:

1. **`getMasterNotifications(String objectName)`**: Returns active rules available for the current object.
2. **`searchRecipients(String type, String searchTerm)`**: Performs a dynamic SOSL/SOQL query based on the selected recipient type (User, Group, or Role) to return Name and ID pairs.
3. **`createSubscription(Id masterId, Id recipientId, String type, Id recordId)`**: Validates and inserts the `WESAL_obj_notificationSubscriber__c` record.

Data Mapping for LWC

When the "Save" action is triggered, the component must map values as follows:

Target Field	Source Value from LWC
`WESAL_fld_masterNotification__c`	Selected Master Notification ID
`WESAL_fld_recipientId__c`	Selected User/Group/Role ID
`WESAL_fld_recipientType__c`	Selected Type (User, Public Group, or Role)
`WESAL_fld_targetRecordId__c`	Current Page `recordId`

User Experience Best Practices

- **Permissions:** The component should only be visible to users with the "WESAL Notification Manager" Permission Set.
- **Validation:** Prevent duplicate subscriptions for the same User/Rule/Record combination using a client-side check before invoking the Apex save method.
- **Search UX:** Implement a minimum 3-character threshold before firing the search query to optimize performance.